

Curriculum Vitae of Kalins Banerjee

Demographic Information

Date of Birth: October 29, 1992

Country of Citizenship: India

Office Address: Department of Urology, University of Michigan, North Campus Research Complex,
2800 Plymouth Rd, Ann Arbor, MI 48109, USA.

Permanent Address: 2/1, Bangur Avenue, Block-A, West Bengal, Kolkata 700055, India.

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Academic Experience

- Research Investigator, Research Faculty Track (starting Sept., 2026)
Division of Urologic Oncology, Department of Urology, University of Michigan, USA.
Adviser: [Professor Evan T. Keller](#)
- Postdoctoral Research Fellow (July, 2025 - Aug., 2026)
Division of Urologic Oncology, Department of Urology, University of Michigan, USA.
Adviser: [Professor Evan T. Keller](#)
- Postdoctoral Research Fellow (Sept., 2021 - June, 2025)
Department of Biostatistics, University of Michigan, USA.
Adviser: [Professor Xiang Zhou](#)

Education

- Ph.D. in Biostatistics (2021), Pennsylvania State University, USA.
Adviser: [Professor Xiang Zhan](#)
Dissertation title: Adaptive Statistical Methods for Microbiome Association Analysis.
- Master of Statistics (2017), Indian Statistical Institute, India.
Specialization: Applied Statistics.
- Bachelor of Science (2015) with Honours in Statistics, Asutosh College, University of Calcutta, India.

Research Interests

- Designing statistical and deep learning methods for genomic and microbiome data.
- Cancer biology, Spatial transcriptomics, copy number variations, and microbiome analysis.

Research Articles [\[Google Scholar\]](#)

1. **Banerjee, K.**, Langefeld, R., Keller, E. T., & Zhou, X. SPICE: A Robust Computational Framework for Identifying Copy Number Variations in Spatial Transcriptomics. *Submitted to **Nucleic Acids Research (NAR)***. [\[R software\]](#)
2. Olson, A. W., Li, J., Li, X. Y., King, L., **Banerjee, K.**, McCoy, A. J., ... & Weiss, S. J. (2026). Amoeboid-Mesenchymal Transition and the Proteolytic Control of Cancer Invasion Plasticity. *Proceedings of the National Academy of Sciences*, 123(15), e2520717123. [\[Paper\]](#)
3. Poulsen, A., Jang, D. H., Khan, M., Al-Mohtaseb, Z. N., Chen, M., **Banerjee, K.**, Scott I. U., & Pantanelli, S. M. (2025). Repeatability of a Dual-Scheimpflug Placido Disc Corneal Tomographer/Topographer in Eyes with Keratoconus. *Clinical Ophthalmology*, 19, 2751–2758. [\[Paper\]](#)
4. Nolan, Z. T., **Banerjee, K.**, Cong, Z., Gettle, S. L., Longenecker, A. L., Kawasaki, Y. I., ... & Nelson, A. M. (2023). Treatment response to isotretinoin correlates with specific shifts in Cutibacterium acnes strain composition within the follicular microbiome. *Experimental Dermatology*, 32(7), 955-964. [\[Paper\]](#)
5. Schneider, A. M., Nolan, Z. T., **Banerjee, K.**, Paine, A. R., Cong, Z., Gettle, S. L., ... & Nelson, A. M. (2023). Evolution of the facial skin microbiome during puberty in normal and acne skin. *Journal of the European Academy of Dermatology and Venereology*, 37(1), 166-175. [\[Paper\]](#) [Journal 2022 IF: 9.2]
6. **Banerjee, K.**, Chen, J., & Zhan, X. (2022). Adaptive and powerful microbiome multivariate association analysis via feature selection. *Nucleic Acids Research (NAR) Genomics and Bioinformatics*, 4(1), lqab120. [\[Paper\]](#) [\[R software\]](#)
[An earlier version won *JSM (Joint Statistical Meetings) 2021 Paper Award* from the **Biometrics** section of American Statistical Association (ASA) ([Announcement](#)) ([Penn State News](#)) and *Best Student Paper Award in Applied Statistics* at the IISA (International Indian Statistical Association) 2021 conference.]
7. Zhan, X., **Banerjee, K.**, & Chen, J. (2021). Variant-Set Association Test for Generalized Linear Mixed Model. *Genetic Epidemiology*, 45(4), 402-412. [\[Paper\]](#) [\[R software\]](#)
8. Kansara, N., Cui, D., **Banerjee, K.**, Landis, Z., Scott, I. U., & Pantanelli, S. M. (2021). Anterior, posterior, and nonkeratometric contributions to refractive astigmatism in pseudophakes. *Journal of Cataract & Refractive Surgery*, 47(1), 93-99. [\[Paper\]](#)
9. Ruzieh, M., Rogers, A. M., **Banerjee, K.**, Soleymani, T., Zhan, X., Foy, A. J., & Peterson, B. R. (2020). Safety of Bariatric Surgery in Patients with Coronary Artery Disease. *Surgery for Obesity and Related Diseases*, 16(12), 2031-2037. [\[Paper\]](#)
10. Schneider, A. M., Cook, L. C., Zhan, X., **Banerjee, K.**, Cong, Z., Imamura-Kawasaki, Y., ... & Nelson, A. M. (2020). Response to Ring et al.: In Silico Predictive Metagenomic Analyses Highlight Key Metabolic Pathways Impacted in the Hidradenitis Suppurativa Skin Microbiome. *The Journal of Investigative Dermatology*, 140(7), 1476-1479. [\[Paper\]](#) [Journal 2019 IF: 7.14]
11. Patel, V. A., Dunklebarger, M., **Banerjee, K.**, Shokri, T., Zhan, X., & Isildak, H. (2020). Surgical Management of Vestibular Schwannoma: Practice Pattern Analysis via NSQIP. *Annals of Otolaryngology, Rhinology & Laryngology*, 129(3), 230-237. [\[Paper\]](#)
12. Schneider, A. M., Cook, L. C., Zhan, X., **Banerjee, K.**, Cong, Z., Imamura-Kawasaki, Y., ... & Nelson, A. M. (2020). Loss of Skin Microbial Diversity and Alteration of Bacterial Metabolic Function in Hidradenitis Suppurativa. *The Journal of Investigative Dermatology*, 140(3), 716-720. [\[Paper\]](#) [Journal 2019 IF: 7.14]

13. **Banerjee, K.**, Zhao, N., Srinivasan, A., Xue, L., Hicks, S. D., Middleton, F. A., ... & Zhan, X. (2019). An adaptive multivariate two-sample test with application to microbiome differential abundance analysis. *Frontiers in genetics*, 10, 350. [\[Paper\]](#) [\[R software\]](#)

[Won a position in the top 5 of the student paper competition at the IISA (International Indian Statistical Association) 2019 conference.]
14. Zeng, Y., Zhu, X., Chen, C., **Banerjee, K.**, Sun, L., Yu, W., ... & Wu, R. (2019). A unified DNA sequence and non-DNA sequence mapping model of complex traits. *The Plant Journal*, 99(4), 784-795. [\[Paper\]](#)

Manuscripts in preparation

1. High definition Spatial Transcriptomic analysis of Breast Cancers with BRCA mutations.
2. The ADAM15/c-Src signaling axis promotes the invasive progression of human Bladder Cancer.
3. Spatial Transcriptomic Profiling of TRIM29-Driven Immune Evasion in Bladder Cancer.

Abstracts

1. Gong, H., The, S., **Banerjee, K.**, Robinson, T., Monda, S., Salami, S. S., Keller, E. T., & May, A. (2026). Abstract A005: Biologic insights from spatial analysis of sarcomatoid renal cell carcinoma metastasis to inform therapeutic strategies. *Cancer Research*, 86(5_Supplement_2), A005. [\[Link\]](#)
2. Nolan, Z. T., **Banerjee, K.**, Cong, Z., Gettle, S., Longenecker, A., Zhan, X., ... & Nelson, A. (2021). Isotretinoin disrupts skin microbiome composition and metabolic function after 20 weeks of therapy. *Journal of Investigative Dermatology*, 141(5), S39. [\[Link\]](#)
3. Peterson, B., **Banerjee, K.**, Zhan, X., & Foy, A. (2019). Patients With Prior Myocardial Infarction or Prior Percutaneous Coronary Intervention Have Increased Perioperative Mortality and Adverse Cardiac Events Following Bariatric Surgery. *Circulation*, 140(Suppl.1), A12242-A12242. [\[Link\]](#)

Commentary on Website

- **Banerjee, K.**, & Keller, E. T. (2025). Cellular annotation: the keystone of spatial analysis. *BioTechniques*. [\[Link\]](#)

Presentations

- “A Robust Computational Framework for Identifying Copy Number Variations in Spatially Resolved Transcriptomics”
Invited, Innovation Across Disciplines Seminar Series (April, 2026), Department of Mathematics and Statistics, South Dakota State University, USA.
- “Robust Inference of Copy Number Variations in Spatial Transcriptomics”
Joint Statistical Meetings (JSM) 2025, USA.
- “An adaptive and powerful multivariate test for microbiome association analysis via feature selection”
 - Joint Statistical Meetings (JSM) 2021, USA.
(Invited as a paper award winner from the Biometrics section).
 - International Indian Statistical Association (IISA) conference 2021.
(Winner of the Student Paper Competition).

- “An adaptive test for microbiome association studies via feature selection”
Eastern North American Region (ENAR) spring meeting 2021, USA.
- “An adaptive multivariate two-sample test with application to microbiome differential abundance analysis”
 - Joint Statistical Meetings (JSM) 2020, USA.
 - International Indian Statistical Association (IISA) conference 2019, India.
(Invited as a finalist of the Student Paper Competition).

Professional Experience

Division of Biostatistics and Bioinformatics, Pennsylvania State University

- Teaching Assistant for Graduate level courses
 - PHS 523: Multivariate Analysis (Fall 2020, online)
Enrollment: Ph.D. students in Biostatistics.
 - PHS 522: Multivariate Biostatistics (Summer 2020, online)
Enrollment: Ph.D., DrPH, MPH, and MD students from various disciplines.
 - PHS 520: Principles of Biostatistics (Fall 2019, online; Fall 2018, in-person)
Enrollment: Ph.D., DrPH, and MD students from various disciplines.
- Research Assistant (Fall 2017 - Spring 2021)

Service

- Session Chair, Senior Ph.D. Student Research Showcase 2023 & 2024, Department of Biostatistics, University of Michigan.
- Judge, JSM 2022 Paper Award Committee, Biometrics section, American Statistical Association.
- Chair, Contributed Session: “Statistical Methods for Large-scale Omics Data”, ENAR 2021.

Reviewing Activities

Nature Biotechnology, Frontiers in Genetics, mSystems.

Computational Skills

R, Python, and SAS.

Awards/Honors

- *JSM (Joint Statistical Meetings) 2021 Paper Award* from the Biometrics section of American Statistical Association (ASA) [[Twitter](#)] [[Penn State News](#)].
- *Best Student Paper Award in Applied Statistics* at the IISA (International Indian Statistical Association) 2021 conference. [[Announcement](#)]
- *Graduate Alumni Society Award* (Penn State College of Medicine, 2020) for outstanding academic and research achievements and contributions to the campus community [[Penn State News](#)].

- *Debesh-Kamal Scholarship* (INR 1 lakh, 2017 - 2018) for the purpose of higher education/research abroad from the Ramakrishna Mission Institute of Culture, Kolkata, India.
- Ranked 2nd in the 3-year Bachelor of Science (B.Sc.) Honours examination (2015), University of Calcutta, India.
Hasi Ganguly Memorial Prize (Asutosh College, 2014 - 2015) for securing highest marks in the B.Sc. final examination in Statistics Honours.
Kesab Lal Ganguly Memorial Prize (Asutosh College, 2013 - 2014) for securing the highest marks in B.Sc. Statistics Honours part-II examination.
- Selected as a member of the 5th delegation from India in the *Japan-East Asia Network of Exchange for Students and Youths Programme (JENESYS)* conducted by the Japan International Cooperation Center (June, 2008).

Contact information for References

- **Evan T. Keller**
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 Southeast University, China.
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 Penn State College of Medicine, Pennsylvania State University, USA.
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- **Seth Pantanelli**
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 Penn State College of Medicine, Pennsylvania State University, USA.
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